How Has Pop Music Changed Over Time

A Data-Driven Journey Through Billboard Hot 100

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Introduction

Popular music reflects culture and societal mood. Songs that reach the Billboard Hot-100 often serve as snapshots of collective emotion, capturing how audiences respond to social, political, and technological change. Over the past two decades, shifts in streaming platforms, social media, and global events have transformed not only how music is consumed, but also how it sounds and feels. This project explores how the emotional tone of popular music has evolved from 2000 to 2023. By using a narrative-driven web experience, this project visualizes trends in musical features associated with emotion and mood, such as danceability and lyrical content, across more than twenty years of chart-topping songs. Rather than presenting static charts, the project employs a scroll-based storytelling format that guides users through key shifts in popular music, allowing them to engage with the data as a cohesive story. The goal is to make abstract changes in musical emotion tangible, accessible, and meaningful to a broad audience.

Motivation

The motivation for this project stems from a personal and cultural curiosity about how popular music has changed. This project seeks to answer this question grounding musical trends in measurable features. The rise of data availability through platforms like Spotify has opened new possibilities for understanding music beyond genre labels or chart rankings. Audio features such as danceability offer quantifiable ways to explore mood and emotional tone at scale. By combining these features with Billboard chart data and lyrics, this project aims to uncover patterns that would not be visible through listening alone. From a storytelling perspective, this project was also motivated by the challenge of moving beyond traditional dashboards and charts. A narrative-driven web experience allows the analysis to unfold gradually, encouraging reflection rather than quick comparison.

Intended Audience

The primary audience for this project includes music enthusiasts, and individuals interested in pop culture trends who may not have a technical background in data science. The visualization is designed to be intuitive and engaging, prioritizing clear annotations, simple interactions, and narrative context over dense statistical explanations. Because the project is web-based, it is easily shareable and accessible across devices, allowing users to explore the story at their own pace.

This flexibility supports a wide range of engagement levels, from quick browsing to deeper exploration of individual years or trends.

Data Source

The dataset used for this project is titled “Billboard Hot-100 [2000–2023] Data with Features”, sourced from Kaggle. It compiles information from three major platforms: Billboard, Genius, and Spotify and includes songs that appeared on the Billboard Hot-100 charts between 2000 and 2023, along with associated lyrics and Spotify audio features. This dataset was selected because it provides a comprehensive and longitudinal view of popular music. Billboard Hot-100 rankings capture mainstream success, while Spotify audio features offer standardized, quantitative measures related to musical characteristics and emotional tone. The inclusion of lyrics further supports qualitative interpretation and future extensions of the analysis.

Data Overview

The dataset contains song-level records with the following relevant fields used in this project:

ranking: The song’s position on the Billboard Hot-100 chart.

song: The name of the track.

band_singer: The performing artist or group.

year: The chart year associated with the song.

lyrics: Full song lyrics sourced from Genius.

danceability: A Spotify audio feature that measures how suitable a track is for dancing, based on tempo, rhythm stability, and beat strength.

Together, these variables allow for analysis of how musical characteristics associated with emotion and mood have shifted over time. For this project, the dataset serves as the foundation for a narrative exploration rather than a predictive or classification task.

Data Preparation and Preprocessing

Before visualization and storytelling, the raw Kaggle dataset was cleaned and transformed to support efficient analysis and frontend rendering. The original dataset was processed into three structured JSON files, each optimized for a specific analytical purpose. First, track-level data was cleaned and standardized to create songs_clean.json, containing one record per song per chart year. Missing or invalid numeric values in Spotify audio features were converted to null and filtered when necessary to prevent distortion of aggregate trends. Only relevant variables required

for storytelling and visualization were retained. Second, year-level aggregates were computed to produce yearly_features.json, which summarizes average audio feature values for each year between 2000 and 2023. These aggregates support long-term trend analysis and enable smooth line and area chart visualizations without requiring repeated computation in the browser. Finally, artist-level summaries were generated in artist_summary.json, aggregating audio features and song counts by artist. This preparation step enables focused artist deep dives that compare individual musical trajectories to broader industry trends. This preprocessing approach balances performance and flexibility. By separating the data into multiple purpose-driven files, the frontend visualizations remain responsive while still allowing for dynamic interaction and narrative-driven transitions. At this stage, the prepared datasets provide a stable foundation for exploratory analysis, visualization development, and iterative refinement of the data-driven story.

Visualization and Narrative approach

This project is designed as a narrative-driven web experience rather than a traditional dashboard. The core storytelling structure follows a scrollytelling format, where the reader progresses through the analysis by scrolling vertically while visualizations update, animate, or transition in response. This approach allows the data story to unfold gradually, guiding the audience through key insights about how popular music has evolved.

The narrative begins with simple, familiar visual forms such as bar charts and line charts to establish historical context and build trust with the audience. As the story progresses, more complex and immersive visualizations are introduced, including multi-dimensional 3D scenes rendered with Three.js. These later sections allow viewers to explore relationships between multiple musical features, such as danceability, energy, and valence, that cannot be easily conveyed through two-dimensional charts alone.Visualizations are tightly synchronized with explanatory text using scroll-based triggers implemented through the Intersection Observer API. This ensures that each visualization appears at the moment it is being discussed, reinforcing the narrative rather than distracting from it. Interactivity is intentionally subtle, prioritizing guided exploration over open-ended manipulation to prevent cognitive overload. Overall, the visualization strategy emphasizes progressive disclosure, moving from overview to detail and from explanation to exploration.

Design Justification

Design decisions throughout the project were guided directly by the evaluation criteria outlined in the project rubric. The story is structured around a clear central question: how has the emotional and sonic character of popular music changed over the past two decades? Each visualization addresses a specific component of this question ensuring that all visual elements contribute meaningfully to the overarching narrative. Creativity and originality are demonstrated through the use of a non-traditional medium. By combining D3.js for conventional 2D charts with Three.js for immersive 3D visualizations, the project moves beyond static representations of data. Elements such as the 3D music space scatter, timeline sphere, and animated waveform transform abstract numerical features into spatial and motion-based metaphors that reflect how music is experienced over time. Technical execution and design quality are reflected in the modular project structure and consistent visual language. Data is preprocessed into clean, purpose-specific JSON files to ensure efficient loading and rendering. Visual encodings such as color, motion, and geometry are consistently mapped to data attributes, reinforcing interpretability. Scroll-based transitions are implemented programmatically rather than manually animated, allowing the experience to remain responsive and scalable. Finally, design choices are justified through a focus on audience engagement and accessibility. Fonts, layout, and spacing prioritize readability, while annotations and callouts provide necessary context for non-technical users. The decision to limit interactivity ensures that the experience remains narrative-first, aligning with the project goal of storytelling rather than exploratory analysis.

Reflection on the storytelling process

Developing this project highlighted the importance of balancing analytical depth with narrative clarity. One of the main challenges was determining how much complexity to expose at each stage of the story. Early iterations included dense visualizations that risked overwhelming the reader. Through iteration, the narrative was refined to introduce concepts incrementally, allowing the audience to build intuition before encountering multi-dimensional views. Another key reflection concerns the use of 3D visualization. While 3D graphics can easily become decorative, this project required careful consideration to ensure that depth, rotation, and animation served an

explanatory purpose. By explicitly tying axes, color gradients, and motion to musical features and time, the 3D elements became tools for understanding rather than spectacle. The scrollytelling format also reinforced the idea that storytelling is as much about pacing as it is about content. Scroll-triggered transitions encouraged a linear reading experience, helping guide interpretation while still allowing moments of exploration. Overall, this project reinforced the value of narrative-driven design in data visualization and demonstrated how combining technical tools with storytelling principles can make complex cultural trends more engaging and accessible.